

Initial release	2014	
Stable release	1.3.6	
Written in	Go	
Licence	Mozilla Public License v2.0	
Website	https://www.terraform.io/	
GitHub repository	https://github.com/hashicorp/terraform	
How is it useful in DevOps practices implementation?	Infrastructure as Code	
Can we integrate it with Pipeline as Code?	Yes. It can be part of Pipeline as Code and different strategies can be used to create resources and deploy applications in different cloud platforms.	
Important Terraform commands	Description	Command
	Verify Terraform version	<code>terraform --version</code>
	Install Terraform modules	<code>terraform init</code>
	Check configuration and create a plan	<code>terraform init -input=false</code>
	Execute the plan	<code>terraform plan</code>
	Delete all resources	<code>terraform apply</code>
	Validate/check the syntax of the Terraform files	<code>terraform apply -auto-approve</code>
	Reformat Terraform configuration in the standard style	<code>terraform destroy</code>
	Update the state to match remote systems	<code>terraform plan --destroy</code>
	Show the current state or a saved plan	<code>terraform validate</code>
	Show the current Terraform version	<code>terraform fmt</code>
	Workspace management	<code>terraform refresh</code>
	Show all commits	<code>terraform show</code>
	Save modified and staged changes	<code>terraform version</code>
	Fetch and merge any commits from the remote branch	<code>terraform workspace</code>
Tutorials	https://learn.hashicorp.com/terraform	

7. Kubernetes

Kubernetes or k8s is an open source, extensible, and portable container orchestration platform founded by Google. It provides declarative configuration and automation for deployment automation. It has become one of the most popular tools in DevOps practices implementation with its

features related to deployment, management, and scaling applications in Kubernetes clusters. It works as a booster in managing resources effectively.

We can create a Kubernetes cluster in an on-premise environment as well as different cloud deployment models such as private and public clouds. Public cloud provides managed services for Kubernetes. Google Cloud Platform has Google Kubernetes Engine, AWS has Amazon Elastic Kubernetes Services (EKS), while Azure has Azure Kubernetes Service (AKS) for managing Kubernetes cluster in a cloud environment.

Some major removals, deprecations and changes in version 1.26 are:

- Based on inclusive naming initiative, the `--prune-whitelist` flag has been replaced with `--prune-allowlist`
- Dynamic kubelet configuration has been removed from the API server
- Deprecations for `kube-apiserver` command line argument `--master-service-namespace`
- Deprecations for `kubectrl run` command line arguments such as `--cascade`, `--filename`, `--force`, `--grace-period`, and so on
- The v1beta1 flow control API group and the v2beta2 HorizontalPodAutoscaler API have been removed

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Written in	Go
Licence	Apache License 2.0
Website	https://kubernetes.io/
GitHub repository	https://github.com/kubernetes/Kubernetes
Features	- Distributed architecture - Easy integration with Docker - Supports hybrid, on-premise, or cloud environments
How is it useful in DevOps practices implementation?	Container orchestration and management. Combined with Infrastructure as Code practice, automation can be achieved in an end-to-end manner for resource provisioning based on the support provided by IaC tool.
Can we integrate it with Pipeline as Code?	Yes
Is a commercial flavour available?	Cloud flavours are available such as EKS, AKS, and GKE